ILLINOIS NCSA

2026 Ashby Prize in Computational Science Hackathon

Welcome & Pre-Briefing · October 12, 2026

Welcome, Hackers!

Two days. Innovation challenge. AI for cancer pathology.

Thank you for your overwhelming interest

- 230 students named at registration
- 60 teams
- 3x larger than expectation and previous 5 hackathons

Two hack days at NCSA: Sat & Sun, Oct 17 - 18

- Work with pretrained pathology models, paired datasets, and campus GPU-compute
- Clinical and computational experts on hand all weekend
- Over $5k in prizes for the top three teams

Who we are

Cancer Center at Illinois (CCIL)
Cancer Digital Insights (CDI) Working Group



Charles Blatti
Research Scientist,
NCSA Visual Analytics



Kun Wang
Assistant Professor,
Comparative Biosciences & BioEngineering

National Center for Supercomputing Applications (NCSA)
Center for Artificial Intelligence Innovation (CAII)



Vlad Kindratenko
Assistant Director,
CAII @ NCSA



Josh Allen
Research Programmer
NCSA Genomics

Who are you and late afternoon energy check

Microsoft Teams Poll

Try 6 questions in less than 3 minutes

Results should appear in Chat



Confirmation Form

If Incomplete, Please Open Now

https://forms.illinois.edu/sec/871865309

Your Challenge

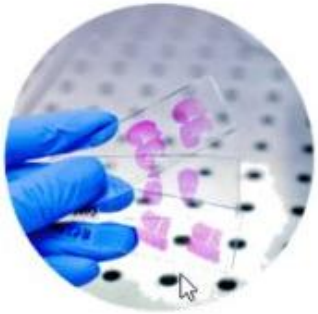
Multi-Modal Pathology Models: Exploring Practical Applications for Improving Clinical Workflows

- Study foundation models trained on pathology images and related medical text
- Leverage and augment them for a specific clinical application your team chooses
- Aim to make clinical pathology more precise, reproducible, or interpretable
- Show working progress and explain why it matters to pathologists and patients



Digital Pathology Workflow

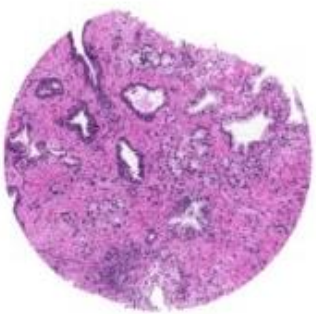
1. Tissue sample



2. Slide Scanning



3. Image Upload



4. Diagnosis

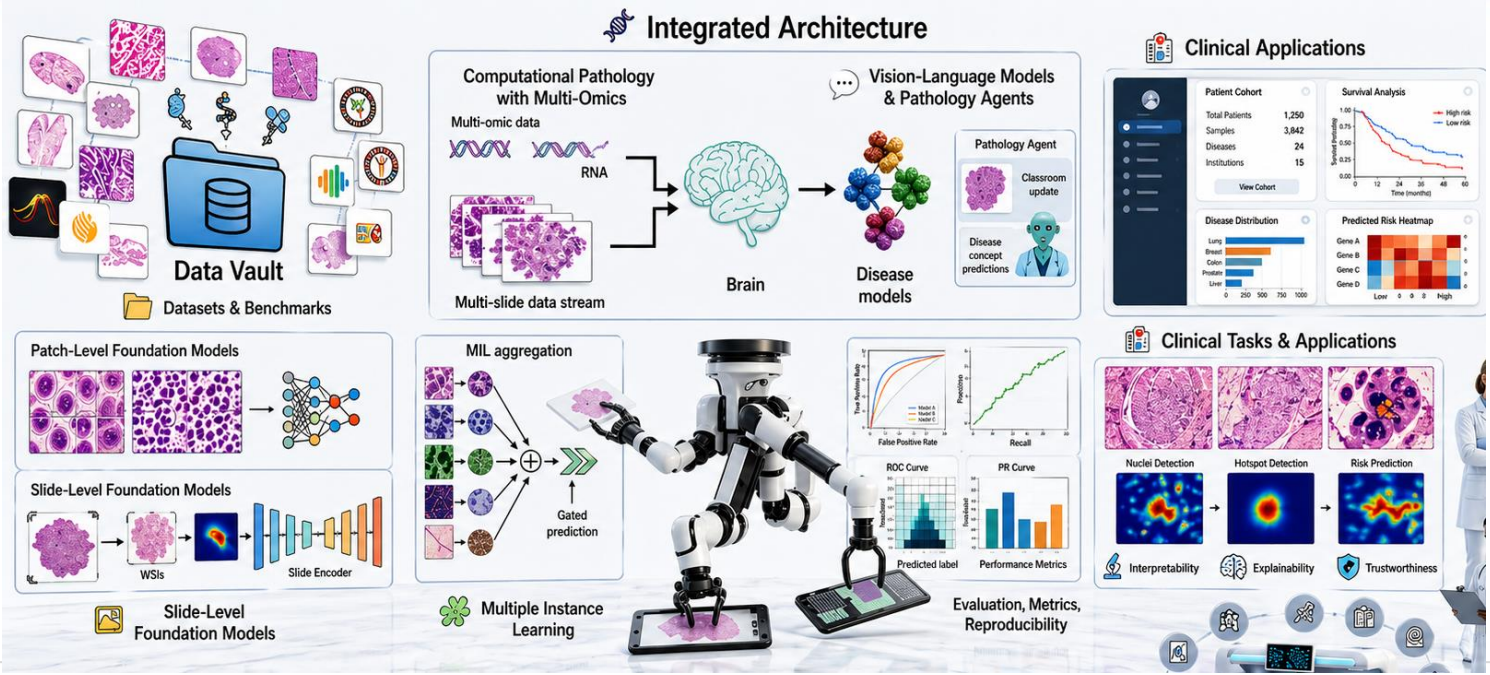


5. Reporting



https://www.pathpresenter.com/digital-pathology-basics-what-is-digital-pathology/

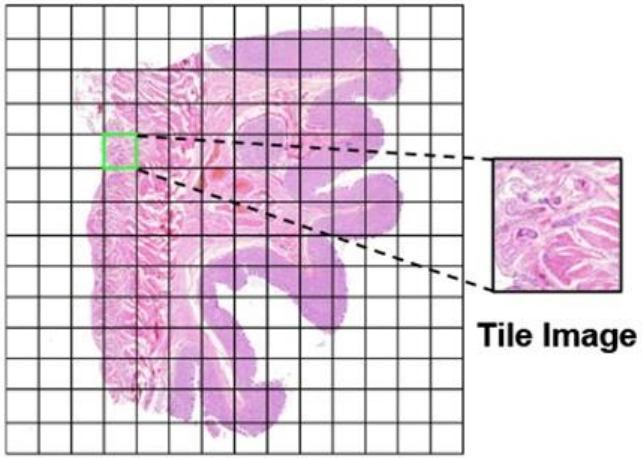
Digital Pathology is a Diverse Field



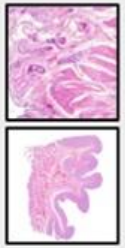
https://github.com/lingxitong/Awesome-AI4DigitalPathology - nearly 500 references

Focus on Tile-Level vs Whole-Slide Image (WSI) Models

Also known as Patch-Level models



Vision-Language Multi-Modality



Tile Caption: Image displays a lymph node featuring a prominent germinal center. This center is enveloped by a mantle zone...

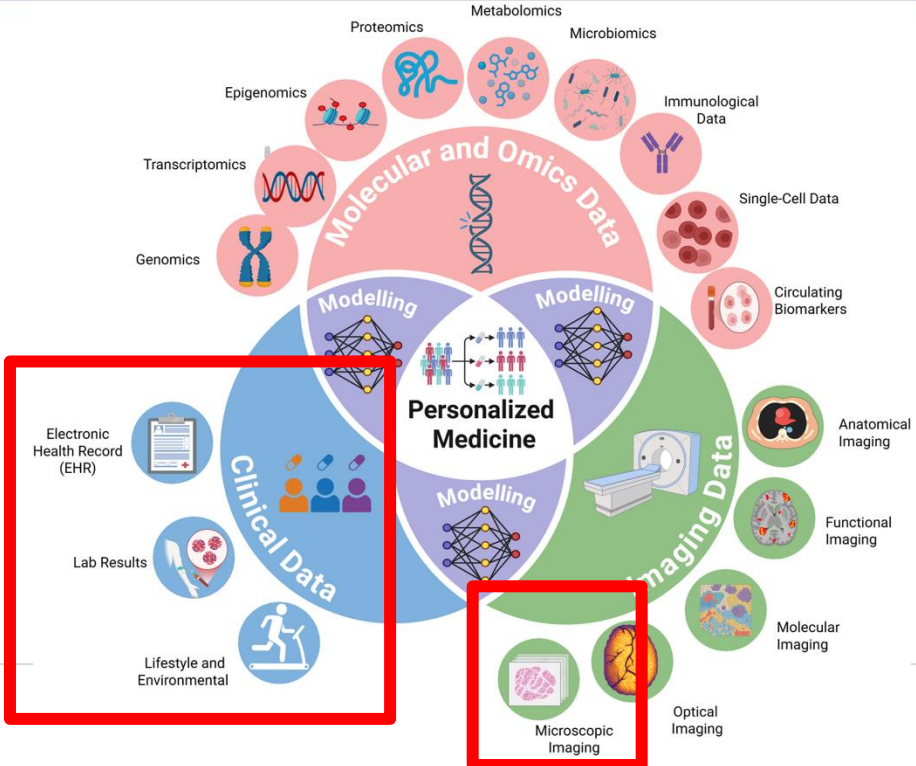
WSI Report: Prostatic adenocarcinoma with a Gleason score of 3 + 4 = 7 (Grade Group 2) confined to the prostate.

https://pmc.ncbi.nlm.nih.gov/articles/PMC13081565/figure/F2/

Vision + Language Foundation Models

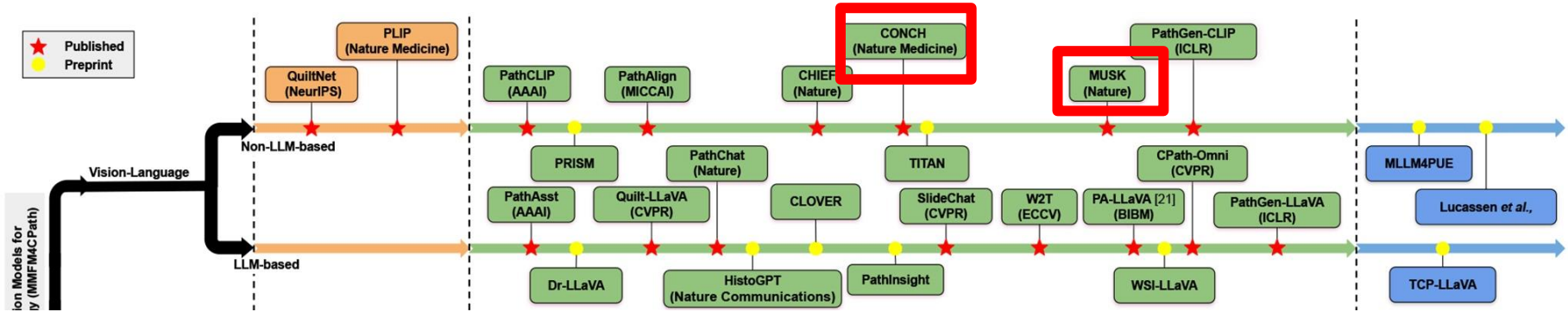
Many different ways to spatially assay a tissue
- Different 2D and 3D imaging assays
- Spatial Omics assays

Focus models that combine 2D microscopic imaging with clinical data text



Muneer, *et al.* From classical machine learning to emerging foundation models: review on multimodal data integration for cancer research. *Artif Intell Rev* (2026). https://doi.org/10.1007/s10462-026-11522-9

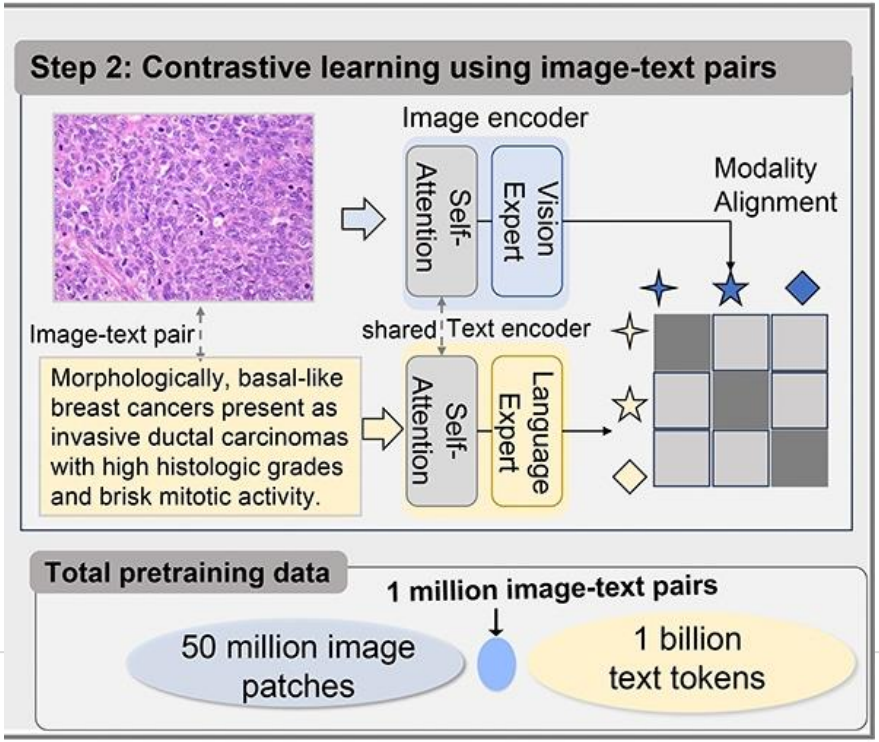
Vision + Language Pathology Foundation Models



https://pmc.ncbi.nlm.nih.gov/articles/PMC13081565/figure/F1/

PreTraining Vision-Language Foundation Models

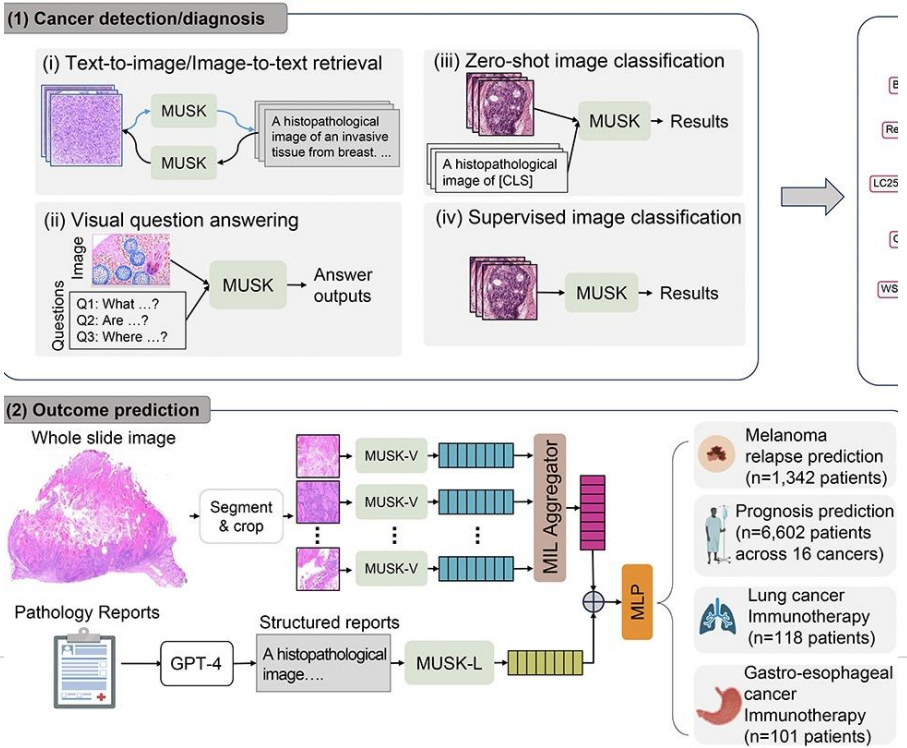
From the MUSK Model



https://pmc.ncbi.nlm.nih.gov/articles/PMC12295649/figure/F1/

Inference tasks with Foundation Models

- Image or Text to Embedding
- Similarity between Image Pairs, Text pairs
- Cross Modal Retrieval
- Zero-Shot Classification of Images or Text
- Visual question answering
- Supervised Classification or prediction
- Image Captioning
- Image segmentation and labeling



From the MUSK Model

https://pmc.ncbi.nlm.nih.gov/articles/PMC12295649/figure/F1/

From Models to Clinical Impact

Pretrained Models

- MUSK
- CONCH
- NCSA hosted open-weight models
- Other models you have access to

Application Datasets

- TCGA slides + reports
- PathMMU image based multiple choice
- Quilt-1M images from Youtube Educational videos
- PathGen combined dataset
- Download or available on NCSA compute

Example Ideas

- Similar case retrieval from image and free-text description
- Report assistant that flags ambiguous terms and suggests improvements
- Education tool that shows sets of example contrasts between two conditions
- Your own clinical question

Hackathon Working Schedule

Revising to accommodate number of participants.

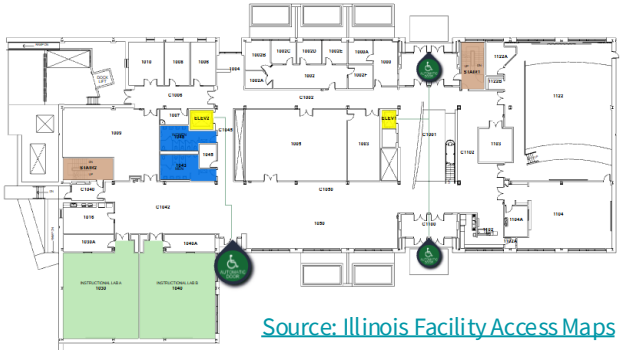
1. **Now:** Kickoff with rules and resources
2. **During the week:** Get access and review materials
3. **Sat Oct 17 & Sun Oct 18:** Hack days. At NCSA from 8:30 am–5:00 pm
4. **Wed Oct 21:** Submit team summaries, presentation ppts, availablity
5. **Oct 22-26:** Judging of project submissions by rubric
6. **Oct TBD:** Top 10* live presentations, awards announced
7. **Nov TBD**: Awards ceremony

Kickoff	Hack Days	Presentations Due
Mon, Oct 12	*Sat–Sun, Oct 17–18*	*Wed, Oct 21*

Limited Rooms at NCSA



- Not enough space at NCSA for all teams
- Find your own team space if you can, otherwise let us know
- Rough room assignments later this week, slack channel for updates
- Staff will be based near main entrances on the first floor



Source: Illinois Facility Access Maps

room_type	room_num	num_seats	num_teams
conference	2000	15	2
conference	2004	8	1
conference	2005	8	1
conference	2013	6	1
conference	2100	20	2
conference	3000	15	2
conference	3004	8	1
conference	3100	20	2
conference	4000	12	2
conference	4042	5	1
conference	4101	12	2
meeting	2001	2	1
meeting	2006	4	1
meeting	2010	4	1
meeting	2036	4	1
meeting	2104	2	1
meeting	2105	2	1
meeting	3010	4	1
meeting	4006	10	1
meeting	4036	4	1
open	1104 lab	40	4
open	1st floor Atrium	50	5
open	2nd floor kitchen	4	1
open	3rd floor kitchen	4	1
open	4th floor kitchen	4	1

Experts and Food at NCSA on Saturday & Sunday

- **8:30 am** Teams start working (light breakfast provided)
- **9:00 am** Orientation meeting on Teams
- **12:00 pm** Lunch (pizza provided)
- **1:00 pm** Teams continue working (snacks provided)
- **4:00 pm** Team briefing on Teams
- **5:00 pm** End of in-person support hours
- NCSA doors will be locked for the weekend, but staff on hand to allow entrance during 830-5pm. Slack channel for building-related issues.

Project Submission

Revised based on number of teams

Initial Submission – by Wed, October 21

- Submit a form with project abstract and team availability for presentations
- Upload a stand-alone presentation (PPT or PDF) describing project aims, methods, results, overcome and remaining challenges, references, etc). No more than 20 main text slides.

Top 10* Live Presentations

- Location and schedule will be posted next week.
- Currently planning 4 teams per hour, 8-minute presentations, 4 minutes of judge questions

Prizes

Over $5K in total awards for the top three teams.

Ashby Prize, established by Steven and Beth Ashby, for transformative potential of high-performance computing in advancing research frontiers.

1st Place	2nd Place	3rd Place
$3,000	**$1,500**	**$750**

Eligibility

Teams strongly encouraged to have at least one member enrolled in the Siebel School of Computing and Data Science

Only current UIUC students are eligible to receive prize money transfers

How You Will Be Judged

- **Innovation & creativity:** novel use of AI, ML, and information science tools or datasets to advance an interesting clinical issue
- **Development progress:** effective use of compute, existing literature, and tools
- **Clinical impact:** relevance and potential effect your solution could have on clinical practice
- **Communication:** quality and clarity of your final summary and oral presentation



Delta and DeltaAI

NCSA's GPU clusters for AI research





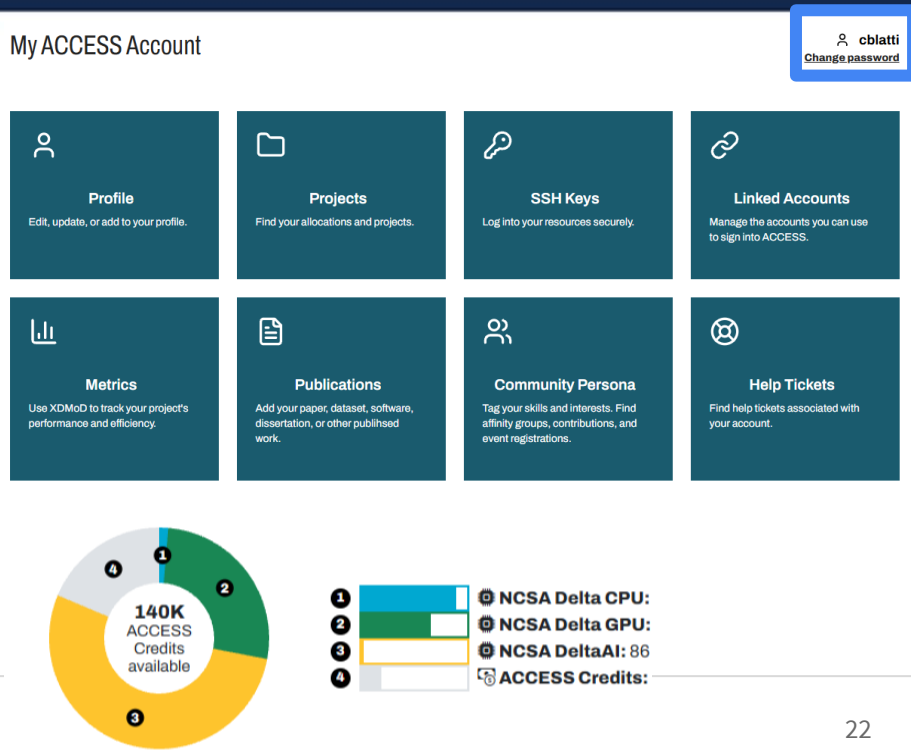
What's Ready for You

- CONCH and MUSK pretrained weights, preloaded
- A few large, paired image–text datasets, preloaded
- GPU nodes for inference, embeddings, and supervised learning
- More instructions later

Using Delta and DeltaAI

Managed through ACCESS

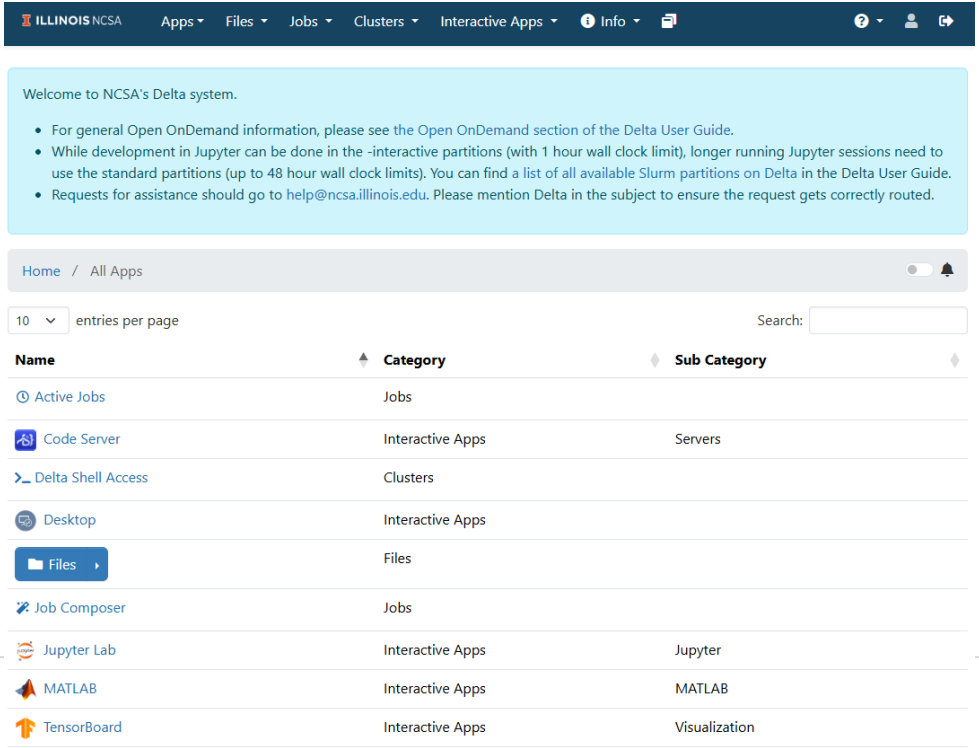
- If you want to have the option of using NCSA compute resources, by **this Wednesday**:
- You **NEED to** [create an ACCESS account] in advance using your institutional email addresses
- Once you have successfully logged into ACCESS, **submit Google form** to get added before the weekend to hackathon project credits account: https://forms.gle/W6fmqQSFroDx6Ccn9



Using Delta and DeltaAI

Documentation

- For information about apps and scheduling on these resources, check out:
- **Delta**: [Documentation][Chat Help]
- **DeltaAI**: [Documentation][Chat Help]



Hugging Face Models and Datasets

Gated Access

- For several models and datasets, you must first ask the owners for access and agree to non-commercial use, including:

https://huggingface.co/xiangjx/musk
https://huggingface.co/MahmoodLab/CONCH
https://huggingface.co/jamessyx/PathGen-CLIP
http://huggingface.co/datasets/jamessyx/PathGen

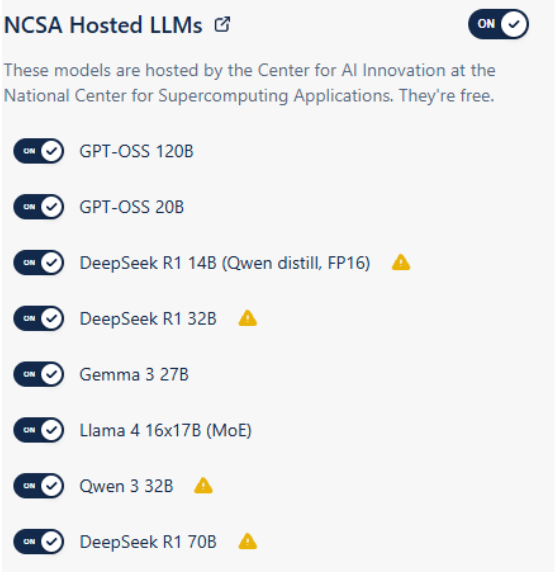
- May take a day or two for request to go through

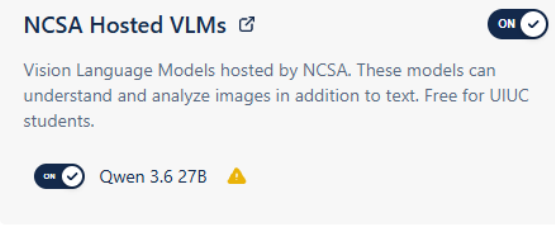


Illinois Chat

Illinois Chat: AI Assistant

- Campus LLM + RAG platform, free with your NetID
- Ask about papers, models, and code
- https://chat.illinois.edu/multimodal-pathology/chat
- Make your own channels and use APIs to access Open Models
- chat.illinois.edu

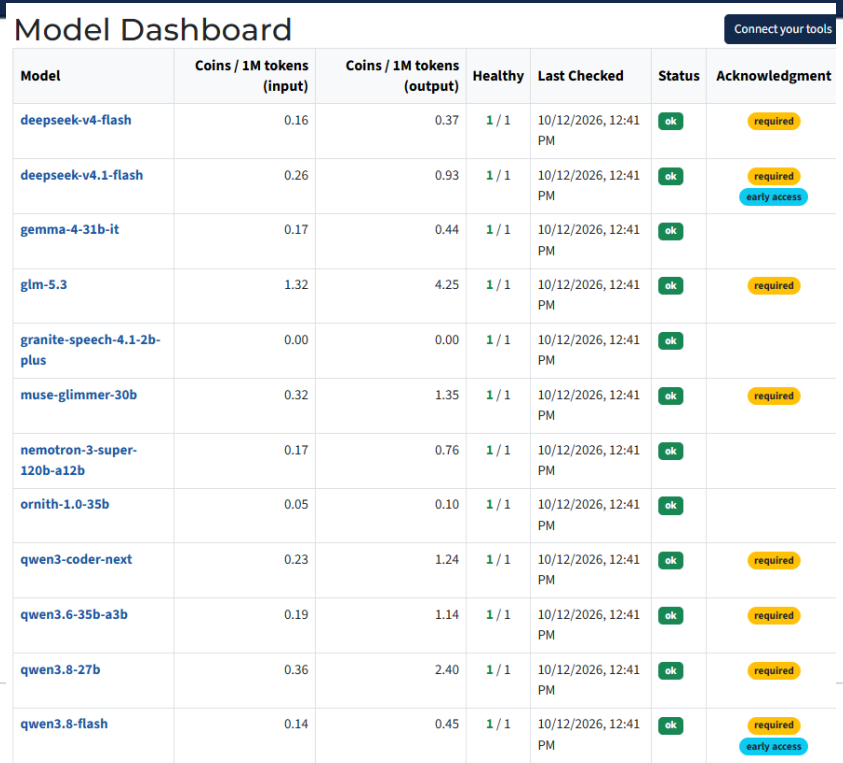




More Local, Open Models

Lumen: AI Model Gateway

- Illinois Computes LLM platform, log in with netid
- Start with 10 coins (usage varies per model by size) and slowly recharges every hour
- Browser-based chat interface or point your own applications to single Lumen API key which can access all models
- **https://lumen.ncsa.illinois.edu/help/**
- Connect your Lumen account to opencode for university resources for agentic coding: Setup Guide



Model Dashboard

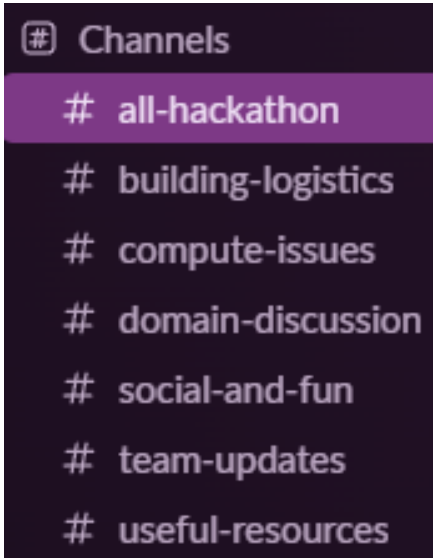
Connect your tools

Model	Coins / 1M tokens (input)	Coins / 1M tokens (output)	Healthy	Last Checked	Status	Acknowledgment
deepseek-v4-flash	0.16	0.37	1 / 1	10/12/2026, 12:41 PM	ok	required
deepseek-v4.1-flash	0.26	0.93	1 / 1	10/12/2026, 12:41 PM	ok	required early access
gemma-4-31b-it	0.17	0.44	1 / 1	10/12/2026, 12:41 PM	ok	
glm-5.3	1.32	4.25	1 / 1	10/12/2026, 12:41 PM	ok	required
granite-speech-4.1-2b-plus	0.00	0.00	1 / 1	10/12/2026, 12:41 PM	ok	
muse-glimmer-30b	0.32	1.35	1 / 1	10/12/2026, 12:41 PM	ok	required
nemotron-3-super-120b-a12b	0.17	0.76	1 / 1	10/12/2026, 12:41 PM	ok	
ornith-1.0-35b	0.05	0.10	1 / 1	10/12/2026, 12:41 PM	ok	
qwen3-coder-next	0.23	1.24	1 / 1	10/12/2026, 12:41 PM	ok	required
qwen3.6-35b-a3b	0.19	1.14	1 / 1	10/12/2026, 12:41 PM	ok	required
qwen3.8-27b	0.36	2.40	1 / 1	10/12/2026, 12:41 PM	ok	required
qwen3.8-flash	0.14	0.45	1 / 1	10/12/2026, 12:41 PM	ok	required early access

First Place to Ask Questions

Slack: Stay Connected

- Announcements and Q&A with organizers and experts
- Can create direct messaging threads for your team
- Harassment and abuse will not be tolerated and result in expulsion
- Slack invite link

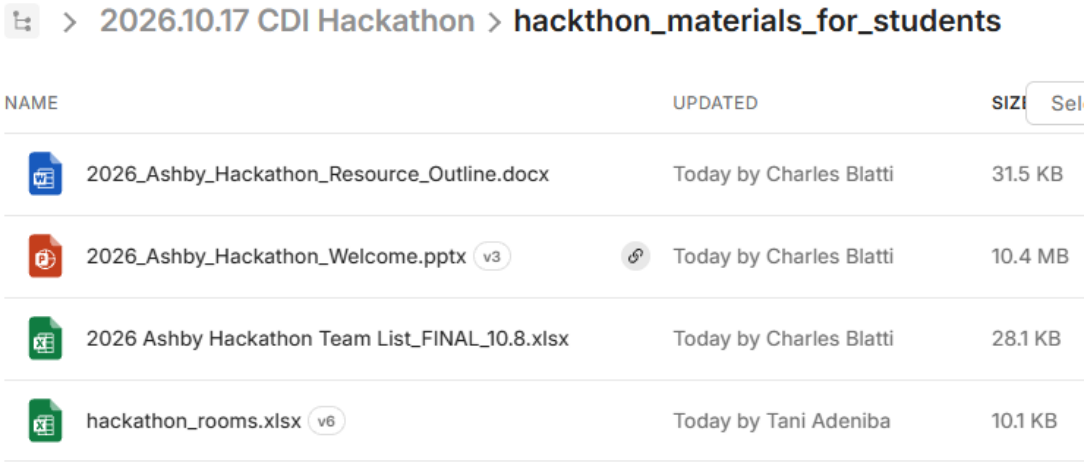


Where to Find Key Info

Box: Shared Files

- Box folder link

- Hackathon Resources
- Updated Team Lists – tomorrow after all confirmations
- NCSA Room Lists
- This Presentation and Recording - soon
- Other Resource Guides - soon



To-Do List for this Week

Arrive Saturday with working access, a team plan, and a first idea.

- **Confirm your participation [by tonight]: Confirmation Form**
- **Create an ACCESS Account [by Wednesday]: ACCESS Registration** and then submit ID to **Google Form**
- Join **Slack**: **Slack invite link**
- Open the **Box** folder to see student materials and pursue a review paper or two: **Box Link**
- **Meet as a team:** choose a clinical question, split roles, sketch a plan
- Explore and try out **Illinois Chat** or **Lumen**
- Request **HuggingFace Access [by Wednesday]** for models and datasets
- We will add you to **Hackathon Compute Account** and let you know **NCSA Room Assignments** by Friday
- **Saturday 8:30 am:** Be ready with your laptop and charger to learn, test ideas, and do something cool

Questions?

Science
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Kun Wang · kwang222@illinois.edu

Technical
Volodymyr Kindratenko · kindrtnk@illinois.edu
Josh Allen · jallen17@illinois.edu

Administrative
Tani Adeniba (CCIL) - tda3@illinois.edu
Pam Joop (NCSA) - pjoop@illinois.edu



Event website

Box Resources

Ask anytime on Slack



blatti@illinois.edu